

The Mental Number Line is Logarithmic Rather than Linear in the Human Intraparietal Sulcus

*Dao Zhou*¹

¹ Department of Psychology and Behavioral Sciences, Zhejiang University, Hangzhou, China

Abstract

The mental number line, the intuitive spatial mapping of numerical magnitudes, has been a cornerstone metaphor in cognitive neuroscience. Yet whether the brain represents numbers on a linear scale (where the distance between 1 and 2 equals that between 7 and 8) or a logarithmically compressed scale (where small numbers are spread apart and large numbers crowded together) remains unresolved. Here we characterize the neural geometry of digits 1–8 in the bilateral intraparietal sulcus (IPS) of five healthy adults from the high-resolution Individual Brain Charting (IBC) dataset. Representational Similarity Analysis (RSA) was employed: we constructed representational dissimilarity matrices (RDM) for each ROI in both single-subject and group average and quantified their correlation with linear versus logarithmic number-line models. Across all subjects, the logarithmic model consistently outperformed the linear model. Strikingly, this logarithmic structure was detectable even in single subjects. Procrustes alignment further revealed significant cross-subject convergence, confirming a shared representational template. These findings provide direct representational evidence that the human IPS encodes number along a logarithmically compressed mental number line, and demonstrate that this architecture constitutes a stable cross-individual cognitive invariant.

Keywords: mental number line, linearity, intraparietal sulcus, logarithmic compression, fMRI, RSA

1. Introduction

Comparing two digits is slower and more error-prone when they are numerically close (e.g., 4 vs. 5) than when they are distant (1 vs. 8). This numerical distance effect (Moyer & Landauer, 1967) reveals that the brain maps numbers onto an ordered continuum, a mental number line (Dehaene, 2003; Restle, 1970). The distance effect only tells us that numbers are ordered. Whether the underlying metric is linear or logarithmic, however, still remains debated. On a linear line, the representational distance between 1 and 2 equals that between 7 and 8. On a logarithmically compressed line, small numbers are broadly spaced while large numbers are crowded together, such that 1 and 2 are as distinct as 4 and 8.

The Weber-Fechner law predicts logarithmic compression: subjective sensation scales logarithmically with physical intensity (Fechner, 1860), and when applied to number, the discriminability of two quantities depends on their ratio rather than their absolute difference. This ratio-dependence has been confirmed across species, from human infants (Xu & Spelke, 2000) to monkeys (Brannon & Terrace, 1998) to rodents (Meck & Church, 1983). Logarithmic scaling can be understood as an efficient neural code: it allocates greater representational bandwidth to small numbers, where fine discrimination is ecologically important, and economizes on large numbers (Gallistel & Gelman, 2000). Yet this view has been challenged by Siegler and Opfer (2003) demonstration that children's explicit number-line estimation shifts from logarithmic to linear with formal education. This logarithmic-to-linear shift raises the possibility that the mature mental number line is linear, and that adult ratio-dependence reflects superficial decision processes rather than the deep representational format. Crucially, however, explicit number-line estimation engages prefrontal and language networks far beyond the core quantity system, and it is possible that the implicit analog magnitude representation retains its logarithmic architecture throughout life while only the explicit behavioral mapping becomes linearized. Resolving this tension requires probing neural population geometry directly.

Single-neuron recordings in macaque monkeys have provided the most direct neural evidence to date. Nieder and Miller (2003, 2004) demonstrated that IPS neurons are tuned to preferred numerosities with Gaussian tuning curves equally spaced on a

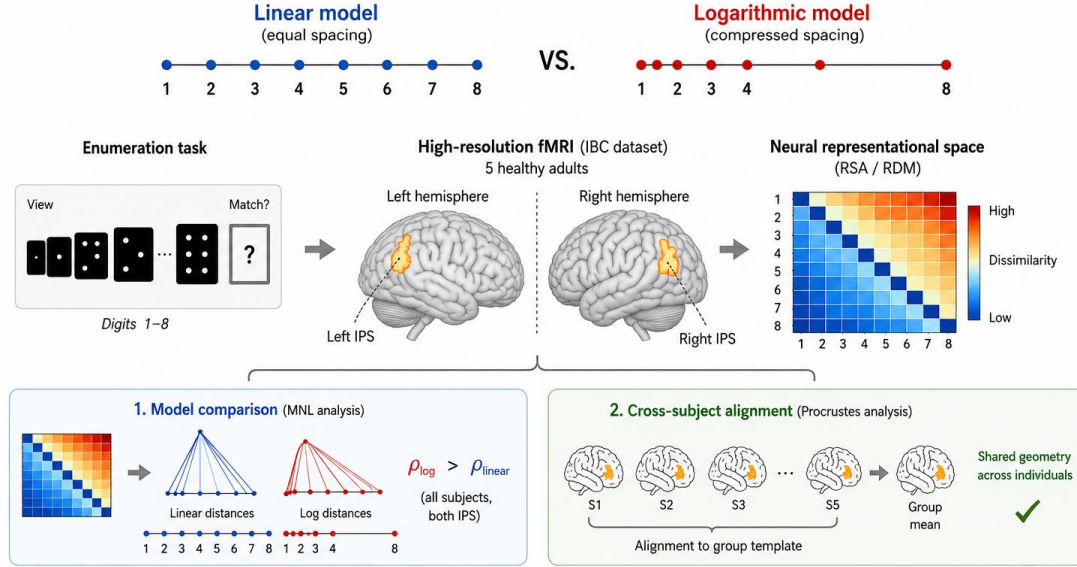


Figure 1. The overview of the analysis pipeline

logarithmic scale, providing an elegant mechanism for Weber’s law: because neurons tuned to larger numbers have broader tuning, the neural code for large numbers is inherently less precise (Nieder & Dehaene, 2009). This finding, however, leaves a critical gap. Single-unit recordings characterize the tuning of individual cells; they do not reveal the population-level representational geometry, the full similarity structure across the distributed code for each number. Whether this logarithmic organization exists at the population level in the human brain, and is accessible with fMRI, remains unknown.

Human neuroimaging has consistently identified the IPS as the core hub for numerical magnitude (Dehaene, 2003; Eger et al., 2009; Piazza et al., 2004), but the vast majority of studies have relied on univariate analyses that can localize where number processing occurs without characterizing how numbers are represented. Representational Similarity Analysis (RSA; Kriegeskorte et al., 2008) addresses this limitation by testing the second-order isomorphism between stimulus similarity and neural pattern similarity, enabling direct, quantitative comparisons between competing representational models. Critically, despite the long-standing log-versus-linear debate, RSA has rarely been applied to this question in individually defined human IPS.

Here we leverage the Individual Brain Charting (IBC) dataset (Pinho et al., 2018; Ponce et al., 2026) to address two questions. First, does IPS representational geometry

follow a linear or a logarithmic metric? We test this by comparing the neural RDM against linear and logarithmic model distance matrices via Spearman correlation with 10,000-label permutation. Second, is this geometry a stable cross-individual invariant? We test this via Procrustes alignment of individual representational spaces. Our results demonstrate that the logarithmic model consistently outperforms the linear model, particularly in the right hemisphere, and that individual geometries significantly converge on a shared template.

2. Methods

2.1 Data

Data were obtained from the Individual Brain Charting (IBC) project (Pinho et al., 2018; Ponce et al., 2026). In IBC, a fixed cohort of twelve participants completed dozens of cognitive tasks in a standardized, high-resolution (1.5 mm isotropic) fMRI acquisition environment over multiple sessions. Unlike conventional group-averaged fMRI studies that acquire limited data per subject and risk obscuring individual representational structure through spatial smoothing and anatomical misalignment, the IBC dataset was designed from the ground up to support within-subject analyses with high statistical power. Five healthy right-handed adults (sub-01, sub-04, sub-05, sub-06, sub-07; 3 female; age 24–38) were included based on the availability of complete Enumeration task data. All participants gave informed consent under protocols approved by the local ethics committee (CPP Île-de-France).

2.2 Experimental Task

Participants viewed displays of 1–8 tilted bars (150 ms), memorized the numerosity, and indicated via button press whether a subsequently presented probe matched (1700 ms). Each trial contained a 1.85-second memorization epoch which was the only epoch modelled in the general linear model. The numerosities spanned both the subitizing range (1–3) and the counting range (4–8). Full protocol details are available in the IBC public repository (<https://individual-brain-charting.github.io/docs/>).

2.3 MRI Acquisition and First-Level GLM

Whole-brain BOLD fMRI data were acquired on a 3T Siemens Prisma scanner (1.5 mm isotropic voxels, TR = 2000 ms, multiband acceleration factor 3) with AP and PA phase-encoding runs. Preprocessing, including motion correction, distortion correction, spatial normalization to MNI152 space, and 3 mm FWHM smoothing, was performed by the IBC consortium.

A first-level GLM was estimated for each subject using SPM12. The design matrix included eight regressors of interest, each modelling the 1.85 s memorization epoch for a specific numerosity convolved with the canonical hemodynamic response function, plus six motion regressors and a 128 s high-pass filter. AP and PA runs were combined within a single fixed-effects model. Contrast images and the average T-contrast map were computed for each subject.

2.4 ROI Definition: AICHA-Based Individual IPS Masking

We defined bilateral IPS regions of interest with a two-stage approach that combines anatomical constraint with functional refinement. In the first stage, the AICHA atlas (Joliot et al., 2015), a whole-brain parcellation of 384 discrete cortical regions, was used to anatomically define the IPS. Labels 109, 111, and 113 (left IPS anterior, middle, and posterior subregions) and labels 110, 112, and 114 (right IPS equivalents) were merged to form bilateral anatomical masks. The AICHA atlas was resampled from its native MNI space (1 mm isotropic) onto each subject's contrast image grid ($105 \times 127 \times 105$ voxels, 1.5 mm isotropic) using nearest-neighbour interpolation. This preserved the discrete label boundaries without partial-volume artifacts and yielded approximately 4,800 anatomical candidate voxels in the left IPS and 5,000 in the right IPS per subject.

In the second stage, we functionally refined these anatomical masks to retain only the most number-responsive voxels. Within each subject's anatomical IPS mask, voxels were ranked by their T-value in the subject's average T-contrast map and the top 10% were retained. This produced functional IPS masks containing 482 voxels in the left hemisphere and 504 voxels in the right hemisphere. Because the average T-map used for voxel selection is orthogonal to the digit-specific pattern information subsequently extracted for RSA, this procedure avoids circularity.

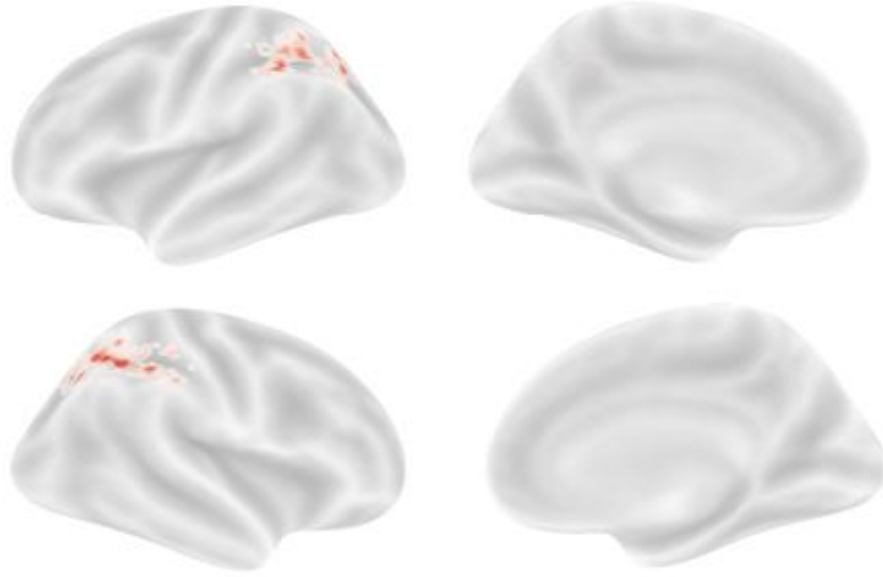


Figure 2. Surface projection of sub-05 IPS functional masks onto fsaverage5 inflated cortex. Four views per subject: left lateral, left medial, right lateral, right medial. Red overlay indicates top-10% functionally-selected voxels

2.5 Representational Similarity Analysis

For each subject and hemisphere, contrast values from con_0001 through con_0008 were extracted from the functional IPS mask. This produced an $8 \times K$ data matrix (K = number of voxels) in which each row represented the K -dimensional activation pattern for a specific digit. Each row was z-scored to zero mean and unit variance, removing overall activation differences across digits while preserving the relative pattern of voxel responses.

An 8×8 representational dissimilarity matrix (RDM) was computed for each subject and hemisphere using correlation distance. The correlation distance between two digits i and j was defined as $d_{ij} = 1 - r(v_i, v_j)$, where r is the Pearson correlation between their z-scored voxel vectors. This yields a symmetric matrix with zeros on the diagonal, where lower values indicate more similar multi-voxel patterns. The group-mean RDM was computed by element-wise averaging across subjects. A group-mean pseudo-subject was also constructed by averaging the z-scored vectors across all five subjects and computing its RDM identically, allowing the group-mean to be included alongside individual subjects in all statistical analyses.

2.6 Matching Number Line Analysis

The central hypothesis test compared the neural RDM against two competing model RDMs. The linear model assumes that neural dissimilarity between two numbers is proportional to their absolute difference: $d_{\text{linear}}(i,j) = |i - j|$. The logarithmic model assumes that neural dissimilarity scales with log-compressed magnitude: $d_{\text{log}}(i,j) = |\log_2(i + 1) - \log_2(j + 1)|$.

For each subject and hemisphere, we extracted the 28 unique pairwise distances from the lower triangle of the 8×8 RDM and computed the Spearman rank correlation with the corresponding distances from each model. Spearman correlation was chosen to avoid assumptions of linearity in the relationship between neural and model distances.

Statistical significance was determined via permutation testing. In each of 10,000 permutations, digit labels were randomly shuffled and model distances were recomputed, while keeping the neural RDM fixed. The exact p-value was computed as $p = (1 + \text{number of permutations where } \rho_{\text{null}} \geq \rho_{\text{observed}}) / (10,000 + 1)$. This tests the null hypothesis that the correspondence between neural and model geometry is no stronger than what would be expected if digits were randomly assigned to neural patterns.

2.7 Procrustes Cross-Subject Alignment

To quantify whether the representational geometry of number is shared across individuals, we performed Procrustes analysis. Because the number of voxels is large relative to the number of digits, direct Procrustes alignment in K -dimensional voxel space would require a singular value decomposition on a $K \times K$ matrix. We therefore first reduced dimensionality via Principal Coordinate Analysis (PCoA). The 8×8 RDM was double-centered, eigendecomposed, and the top seven eigenvectors ($n - 1$ for the eight-digit space) were retained and scaled by the square root of their eigenvalues. This produced an 8×7 coordinate matrix for each subject and hemisphere, representing the digit geometry in a low-dimensional Euclidean space that optimally preserves the correlation-distance relationships in the original RDM.

Procrustes transformation (optimal rotation, scaling, and reflection, without shearing or non-linear warping) was then applied to align subject pairs and to align each

subject to the group-mean template. The residual sum of squares after alignment, the Procrustes disparity, quantifies the geometric deviation between two representational spaces. Each disparity was tested against a null distribution generated by 10,000 random row-shuffles of the target matrix.

3. Results

3.1 The Mental Number Line is Logarithmic Rather Than Linear

The MNL analysis revealed three converging lines of evidence supporting a log-compressed mental number line.

The logarithmic model outperformed the linear model in every single comparison. Across all 12 subject-by-hemisphere conditions, $\rho_{\log}$ exceeded ρ_{linear} without exception (binomial $p < 0.001$). The mean advantage of the logarithmic model was $\Delta\rho = +0.210$ in the right IPS and $+0.157$ in the left IPS.

The logarithmic model achieved statistical significance in cases where the linear model did not. Four of the five individual subjects showed significant logarithmic correlations ($p < 0.05$) in the right IPS; only one subject reached significance with the linear model. The group-mean results were particularly pronounced. The right IPS group-mean achieved $\rho_{\log} = 0.679$ ($p < 0.0001$), nearly double the linear ρ of 0.387. In the left IPS, the group-mean $\rho_{\log}$ was 0.525 ($p < 0.001$), more than double the non-significant linear ρ of 0.220.

The right hemisphere consistently showed stronger logarithmic coding than the left. Every subject exhibited a higher logarithmic ρ in right IPS than in left IPS. Four of five subjects also showed higher linear ρ in the right hemisphere.

A notable finding was that the logarithmic structure was already detectable at the single-subject level. While group averaging improved the signal-to-noise ratio as expected (the group-mean $\rho_{\log}$ exceeded every individual subject's $\rho_{\log}$), the fact that four out of five individual subjects independently yielded statistically significant logarithmic correlations indicates that the effect is robust and does not depend on averaging. This pattern, consistent single-subject significance combined with group-level

enhancement, is precisely what one would expect from a shared representational architecture expressed through individually noisy neural implementations.

Table 1. MNL Spearman correlations between IPS RDMs and number-line models.

Subject	ROI	ρ_{linear}	p	ρ_{log}	p	$\Delta (\text{log} - \text{linear})$
sub-01	ips_left	0.024	ns	0.173	ns	+0.149
sub-01	ips_right	0.123	ns	0.243	ns	+0.120
sub-04	ips_left	0.346	*	0.456	**	+0.110
sub-04	ips_right	0.387	*	0.655	***	+0.268
sub-05	ips_left	0.100	ns	0.358	*	+0.258
sub-05	ips_right	0.319	~	0.566	***	+0.247
sub-06	ips_left	0.207	ns	0.461	**	+0.254
sub-06	ips_right	0.312	~	0.577	***	+0.265
sub-07	ips_left	0.282	~	0.291	~	+0.009
sub-07	ips_right	0.346	*	0.437	*	+0.091
group-mean	ips_left	0.220	ns	0.525	***	+0.305
group-mean	ips_right	0.387	*	0.679	***	+0.292

$p < 0.001$, $p < 0.01$, $p < 0.05$, $\sim p < 0.10$, ns = not significant. All p -values from 10,000-label permutation test. *

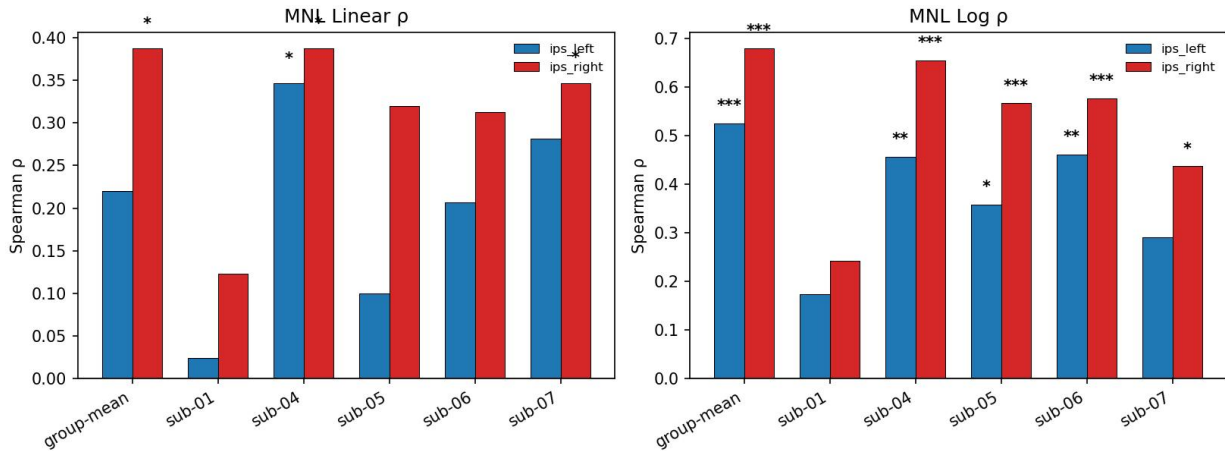


Figure 3. MNL Spearman correlations between IPS RDMs and number-line models.

3.2 Cross-Subject Geometric Stability

Procrustes analysis quantifies the maximal geometric similarity between two representational spaces after optimal rotation, scaling, and reflection. We used this to compare the number representation manifold across subjects and between each individual subject and the group mean.

In the right IPS, four of five subjects showed significant alignment to the group template ($p < 0.05$), and three of five showed significant alignment in the left IPS. Subject sub-05 exhibited the strongest convergence to the group (disparities of 0.075–0.105), while sub-06 showed the weakest (0.227–0.304), suggesting greater individual deviation in this subject’s number geometry.

Pairwise subject-to-subject comparisons, however, yielded fewer significant results. Only three of fifteen right-IPS pairs and three of fifteen left-IPS pairs were individually significant. This is likely because the signal-to-noise ratio in individual subjects was relatively modest relative to the sensitivity of the pairwise Procrustes test. The logic is straightforward. Comparing two noisy individual representations directly compounds the noise from both. Comparing each individual to the group template, which averages out idiosyncratic noise, allows the shared structure to emerge more clearly. The fact that most subjects aligned significantly to the template while few subject pairs aligned to each other suggests that individual geometries do converge on a common logarithmic scaffold, but that each subject’s implementation carries sufficient idiosyncratic variation to obscure this convergence in direct pairwise comparisons.

Table 2. Procrustes alignment of individual subjects to the group-mean template.

Subject	Left IPS disparity	p	Right IPS disparity	p
sub-01	0.143	**	0.101	**
sub-04	0.139	ns	0.101	**
sub-05	0.105	***	0.075	***
sub-06	0.227	*	0.304	*
sub-07	0.096	ns	0.090	~

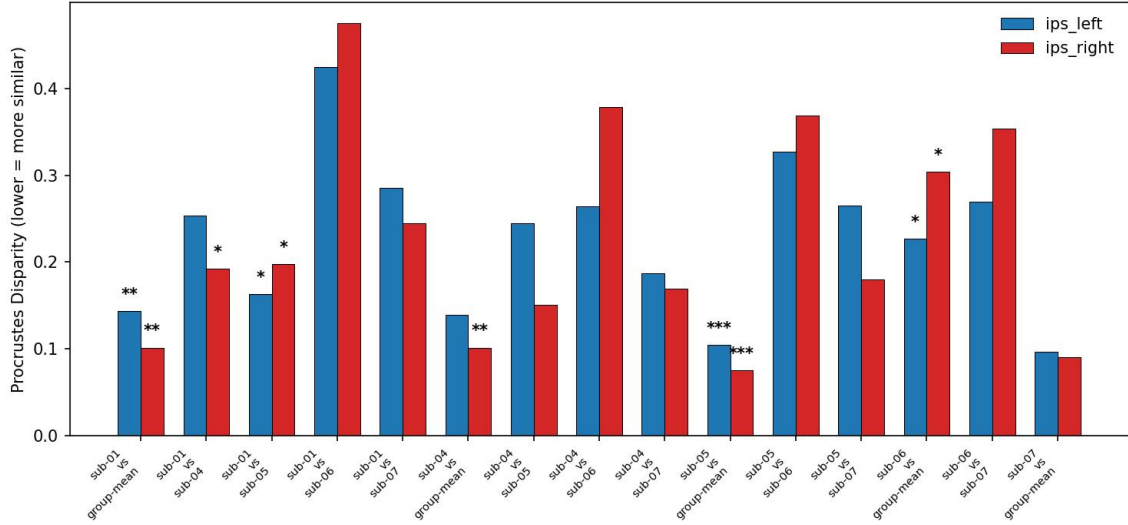


Figure 4. Procrustes disparity bar chart showing alignment for each subject versus the group-mean template, separately for left and right IPS. Lower bars indicate greater geometric similarity. Significance stars ($p<0.05$, $p<0.01$, $p<0.001$) reflect 10,000-row-shuffle permutation tests.

4. Discussion

We found that the representational geometry of digits in the human IPS is better described by a logarithmic metric than a linear one, and that this geometry is shared across individuals. Below we discuss the implications of these findings.

4.1 Logarithmic Compression as a Neural Design Principle

The logarithmic organization we observed provides a direct neural instantiation of the Weber-Fechner law at the representational level. Previous fMRI work demonstrated Weber-like behavior in IPS activation amplitude (Piazza et al., 2004) but did not characterize the similarity structure of the population code. Our RSA results go further by showing that the multi-voxel pattern for a digit is more similar to the pattern of its numerical neighbor when that neighbor is a large number than when it is a small number. Digits 1 and 2 (ratio 2:1) are as representationally distinct as digits 4 and 8 (also ratio 2:1), despite vastly different absolute differences. This is the population-level signature of a log-compressed code.

This finding converges with single-neuron recordings in macaque monkeys. Nieder and Miller (2003, 2004) showed that IPS neurons are tuned to preferred numerosities with Gaussian tuning curves equally spaced on a logarithmic axis. The representational geometry we observe, where numerical ratio rather than absolute difference governs similarity, is precisely what one would predict from a population of such logarithmically tuned neurons. This cross-species convergence strengthens the case for logarithmic scaling as a fundamental, evolutionarily conserved design principle. It makes functional sense: logarithmic coding allocates greater representational precision to small numbers, where fine discrimination matters for fitness, while allowing the system to represent very large numbers without requiring proportionally more neural resources.

4.2 Reconciling the Logarithmic-to-Linear Shift with Adult Logarithmic Coding

Our findings also speak to the tension between psychophysical evidence for logarithmic representation and the developmental logarithmic-to-linear shift documented by Siegler and Opfer (2003). We suggest that these findings are not contradictory but reflect two distinct levels of representation. The implicit analog magnitude system in IPS, tapped by rapid non-symbolic quantity perception in our Enumeration task, may remain logarithmically organized throughout life. The explicit, culturally mediated mapping of numerals onto space, tapped by paper-and-pencil number-line estimation, becomes linearized through education. This dual-representation account is consistent with the Triple Code Model (Dehaene & Cohen, 1995), which posits distinct analog magnitude, verbal, and visual digit codes in the parietal lobe. Under this framework, our RSA analysis of the Enumeration task selectively probes the analog magnitude code, which retains its evolutionary logarithmic signature, while behavioral number-line estimation tasks additionally engage the verbal and visual codes that support linear mapping strategies.

4.3 Right-Hemisphere Dominance and the ATOM Framework

The consistent right-hemisphere advantage we observed aligns with a substantial body of evidence. Right parietal lesions are more likely to produce numerical deficits than left parietal lesions (Cipolotti et al., 1991). TMS over right IPS disrupts quantity

processing more than left IPS stimulation (Cappelletti et al., 2007). Meta-analyses consistently find stronger right IPS activation for numerical tasks (Arsalidou & Taylor, 2011). One influential interpretation is Walsh's (2003). A Theory of Magnitude, which proposes that right parietal cortex is specialized for magnitude processing broadly construed, encompassing number, time, space, and size, while left parietal regions are more involved in verbal and symbolic processing. Our findings are consistent with this framework: every subject in our study showed stronger logarithmic coding in the right IPS than the left, suggesting that the right hemisphere is the primary locus of the logarithmically compressed analog magnitude code.

4.4 Cross-Subject Stability and Individual Variability

The cross-subject convergence revealed by Procrustes analysis, combined with the presence of significant individual variation, reflects a broader principle familiar from other cognitive domains. Cortical networks for face and language processing show highly consistent gross localization across individuals but variable fine-scale functional organization. The mental number line appears to follow the same pattern. A logarithmic representational scaffold is shared across brains, likely reflecting a species-typical cognitive architecture, but the precise implementation allows for individual differences shaped by genetics, experience, and education. The fact that four of five subjects independently yielded significant logarithmic MNL correlations is notable. It demonstrates that the effect is not an artifact of group averaging but a genuine property of individual brains, detectable despite the inherent noise of single-subject fMRI.

4.5 Limitations

Several limitations should be noted. Our sample of five subjects, while typical for high-resolution within-subject fMRI designs, limits the generalizability of our findings. Replication in the full IBC cohort ($N = 12$) and in independent datasets would be valuable, though the consistency of the within-subject effects (log outperformed linear in all 12 comparisons; right exceeded left in every subject) provides confidence that the results are not driven by outliers. The numerical range of 1–8 straddles the subitizing-to-counting boundary, and it is possible that representational geometry differs between these

ranges. Testing larger numerosities, where logarithmic compression becomes more pronounced, would provide an even stronger test of the logarithmic hypothesis. Our analysis was restricted to a single task and a single ROI. The IBC dataset includes multiple numerical tasks in the same subjects, and extending the present analysis across tasks would address whether the logarithmic geometry we observed is task-specific or generalizes across numerical contexts. Finally, our model comparison was limited to two candidate models (linear and logarithmic). Other representational formats, such as power-law compression or piecewise linear scaling, may provide better fits, and formal Bayesian model comparison would enable more rigorous adjudication among competing accounts.

5 Conclusion

The neural geometry of number in the human IPS follows a logarithmically compressed metric. This architecture bridges classical psychophysics with modern population-level neural geometry, converges with single-neuron recordings in non-human primates, and is conserved across individuals as a stable cognitive invariant, while also exhibiting quantifiable individual variation. The mental number line is not a ruler. It is a logarithmic slide rule, etched into the population codes of the human parietal cortex.

References

- Arsalidou, M., & Taylor, M. J. (2011). Is $2+2=4$? Meta-analyses of brain areas needed for numbers and calculations. *NeuroImage*, 54(3), 2382–2393. <https://doi.org/10.1016/j.neuroimage.2010.10.009>
- Brannon, E. M., & Terrace, H. S. (1998). Ordering of the numerosities 1 to 9 by monkeys. *Science*, 282(5389), 746–749. <https://doi.org/10.1126/science.282.5389.746>
- Cappelletti, M., Barth, H., Fregni, F., Spelke, E. S., & Pascual-Leone, A. (2007). rTMS over the intraparietal sulcus disrupts numerosity processing. *Experimental Brain Research*, 179(4), 631–642. <https://doi.org/10.1007/s00221-006-0816-0>
- Cipolotti, L., Butterworth, B., & Denes, G. (1991). A specific deficit for numbers in a case of dense acalculia. *Brain*, 114(6), 2619–2637. <https://doi.org/10.1093/brain/114.6.2619>

- Dehaene, S. (2003). The neural basis of the weber–fechner law: A logarithmic mental number line. *Trends in Cognitive Sciences*, 7(4), 145–147.
[https://doi.org/10.1016/S1364-6613\(03\)00055-X](https://doi.org/10.1016/S1364-6613(03)00055-X)
- Dehaene, S., & Cohen, L. (1995). Towards an anatomical and functional model of number processing. *Mathematical Cognition*, 1(1), 83–120.
- Eger, E., Michel, V., Thirion, B., Amadon, A., Dehaene, S., & Kleinschmidt, A. (2009). Deciphering cortical number coding from human brain activity patterns. *Current Biology*, 19(19), 1608–1615. <https://doi.org/10.1016/j.cub.2009.08.047>
- Fechner, G. T. (1860). *Elemente der psychophysik*. Breitkopf und Härtel.
- Gallistel, C. R., & Gelman, R. (2000). Non-verbal numerical cognition: From reals to integers. *Trends in Cognitive Sciences*, 4(2), 59–65.
[https://doi.org/10.1016/S1364-6613\(99\)01424-2](https://doi.org/10.1016/S1364-6613(99)01424-2)
- Joliot, M., Jobard, G., Naveau, M., Delcroix, N., Petit, L., Zago, L., Crivello, F., Mellet, E., Mazoyer, B., & Tzourio-Mazoyer, N. (2015). AICHA: An atlas of intrinsic connectivity of homotopic areas. *Journal of Neuroscience Methods*, 254, 46–59.
<https://doi.org/10.1016/j.jneumeth.2015.07.013>
- Kriegeskorte, N., Mur, M., & Bandettini, P. A. (2008). Representational similarity analysis: Connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 4. <https://doi.org/10.3389/neuro.01.004.2008>
- Meck, W. H., & Church, R. M. (1983). A mode control model of counting and timing processes. *Journal of Experimental Psychology: Animal Behavior Processes*, 9(3), 320–334. <https://doi.org/10.1037/0097-7403.9.3.320>
- Moyer, R. S., & Landauer, T. K. (1967). Time required for judgements of numerical inequality. *Nature*, 215(5109), 1519–1520. <https://doi.org/10.1038/2151519a0>
- Nieder, A., & Dehaene, S. (2009). Representation of number in the brain. *Annual Review of Neuroscience*, 32(1), 185–208.
<https://doi.org/10.1146/annurev.neuro.051508.135550>
- Nieder, A., & Miller, E. K. (2003). Coding of cognitive magnitude: Compressed scaling of numerical information in the primate prefrontal cortex. *Neuron*, 37(1), 149–157.
[https://doi.org/10.1016/S0896-6273\(02\)01177-6](https://doi.org/10.1016/S0896-6273(02)01177-6)
- Nieder, A., & Miller, E. K. (2004). A parieto-frontal network for visual numerical information in the monkey. *Proceedings of the National Academy of Sciences*, 101(19), 7457–7462. <https://doi.org/10.1073/pnas.0402238101>
- Piazza, M., Izard, V., Pinel, P., Le Bihan, D., & Dehaene, S. (2004). Tuning curves for approximate numerosity in the human intraparietal sulcus. *Neuron*, 44(3), 547–555. <https://doi.org/10.1016/j.neuron.2004.10.014>

- Pinho, A. L., Amadon, A., Ruest, T., Fabre, M., Dohmatob, E., Denghien, I., Ginisty, C., Becuwe-Desmidt, S., Roger, S., Laurier, L., Joly-Testault, V., Médiouni-Cloarec, G., Doublé, C., Martins, B., Pinel, P., Eger, E., Varoquaux, G., Pallier, C., Dehaene, S., ... Thirion, B. (2018). Individual brain charting, a high-resolution fMRI dataset for cognitive mapping. *Scientific Data*, 5(1), 180105. <https://doi.org/10.1038/sdata.2018.105>
- Ponce, A. F., Aggarwal, H., Shankar, S., Torre, J. J., Pinho, A. L., Thual, A., Ginisty, C., Lecomte, Y., Berland, V., Beriot, L., Laurier, L., Joly-Testault, V., Médiouni-Cloarec, G., Hertz-Pannier, L., Doublé, C., Martins, B., Amalric, M., Dehaene, S., Diersch, N., ... Thirion, B. (2026). Individual Brain Charting: Fifth release of high-resolution fMRI data for cognitive mapping. *Scientific Data*, 13(1), 593. <https://doi.org/10.1038/s41597-026-06869-1>
- Restle, F. (1970). Speed of adding and comparing numbers. *Journal of Experimental Psychology*, 83(2, Pt.1), 274–278. <https://doi.org/10.1037/h0028573>
- Siegler, R. S., & Opfer, J. E. (2003). The development of numerical estimation: Evidence for multiple representations of numerical quantity. *Psychological Science*, 14(3), 237–250. <https://doi.org/10.1111/1467-9280.02438>
- Walsh, V. (2003). A theory of magnitude: Common cortical metrics of time, space and quantity. *Trends in Cognitive Sciences*, 7(11), 483–488. <https://doi.org/10.1016/j.tics.2003.09.002>
- Xu, F., & Spelke, E. S. (2000). Large number discrimination in 6-month-old infants. *Cognition*, 74(1), B1–B11. [https://doi.org/10.1016/S0010-0277\(99\)00066-9](https://doi.org/10.1016/S0010-0277(99)00066-9)